Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 125 (2018) 234–240

Procedia Computer Science

www.elsevier.com/locate/procedia

6th International Conference on Smart Computing and Communications, ICSCC 2017, 7-8 December 2017, Kurukshetra, India

Improved Cuckoo Search-based Clustering Protocol for Wireless Sensor Networks

Govind P. Gupta*

Department of Information Technology
National Institute of Technology, Raipur 492010, India

Abstract

In a large-scale Wireless Sensor Networks (*WSNs*), designing of an energy efficient data gathering protocol has become a challenging research issues. This is due to fact that each sensor node is generally equipped with limited energy resource. In the literature, clustering-based technique for data gathering has been proved very efficient in terms of energy saving. Although, it is pointed out in the literature that design of an energy-balanced clustering for maximizing the network lifetime of WSNs is a NP-hard problem. For solving this NP-hard problem, many meta-heuristic approach based clustering protocols are proposed in the recent years. However, these existing clustering protocols suffer from unbalanced energy consumption problem. In this problem, cluster heads are not uniformly distributed and overloaded cluster heads die out faster than under-load cluster heads. In order to solve this problem, an improved Cuckoo Search-based Clustering Algorithm (*ICSCA*) is proposed in this research paper. Performance evaluation of the *ICSCA* and its comparison with the state-of-art clustering scheme in terms total energy consumption and residual energy are presented.

© 2018 The Authors. Published by Elsevier B.V.
Peer-review under responsibility of the scientific committee of the 6th International Conference on Smart Computing and Communications.

Keywords: Wireless sensor networks, Clustering, Cuckoo Search

* Corresponding author. Tel.: +91-9458127005.
E-mail address: gpgupta.it@nitrr.ac.in

1877-0509 © 2018 The Authors. Published by Elsevier B.V.
Peer-review under responsibility of the scientific committee of the 6th International Conference on Smart Computing and Communications
10.1016/j.procs.2017.12.032

1. Introduction

Advancement in wireless communication and VLSI technologies motivates the manufacturers to design low cost and small size wireless sensor devices that can be used in the design of wireless sensor networks (*WSNs*) [1, 2]. A *WSN* is a distributed network which contains a set of autonomous and self organised sensor nodes and one or more Base Station (*BS*). This kind of network can be designed and deployed for monitoring and tracking applications such as smart environments like smart cities, smart grid, smart home; habitat monitoring, military target tracking and surveillance etc [1, 3].

In WSNs, a sensor node is equipped with limited energy resource. Generally, it is operated using 2AA batteries. Due to the limitation of energy resources, judicial uses of the energy resources of the sensor node is an important research issue which required urgent research effort for better solution in order to enhance the network lifetime. For saving of the energy resource of the sensor node and enhancing the network lifetime, various methods are proposed such as clustering, data compression and aggregation, mobile sink based data gathering, duty cycling etc. Clustering based data gathering scheme has been proved in many literature for providing energy efficient and scalable solutions [4-6].

In clustering process, nodes are organized into various groups known as clusters. Each cluster is provided with a Cluster Head (*CH*) whose main work is to receive the sensed data from its cluster members (*CMs*), aggregate it and then transport the aggregated data to the base station [4-6]. In each cluster, *CH* removes redundant data by doing data aggregation. Thus, proper selection of *CHs* and its spatial distribution are very important issues in the energy balanced clustering process.

The problem of energy balanced cluster head selection is NP-hard problem. For solving this problem, many heuristics and meta-heuristic based clustering algorithms are proposed in the literature [6-14]. However, these existing clustering protocols are suffering from unbalanced energy consumption problem. In this problem, *CHs* are not uniformly distributed and overloaded *CHs* die-out faster than under-load *CHs*. This is due to fact that average distance between *CH* and *BS* is used in the fitness function which causes selection of all *CHs* near to the *BS*. In order to solve this problem, an improved Cuckoo Search [15-17] based Clustering Algorithm (*ICSCA*) is proposed. In the proposed scheme, a novel fitness function is derived which includes three parameters such as energy, distance and cluster size. For improving the performance of Cuckoo Search, a novel encoding scheme is used for encoding the population. Performance evaluation of the proposed clustering scheme and its comparison with the state-of-art clustering scheme such as LEACH [3], E_OEERP [12], and PSO-ECHS [13] are discussed.

The remaining section of the paper is organized as follows: Section 2 discusses related work on the meta-heuristic optimization-based clustering scheme. Section 3 presents the proposed clustering scheme. In Section 4, performance evaluation of the proposed scheme and its comparison with the well known clustering scheme are included. Finally Section 5 concludes the proposed work.

2. Related Works

Clustering protocols are categorized into two groups: heuristic-based clustering and nature-inspired computing-based clustering [6]. Several heuristic protocols have been developed for increasing the lifetime of sensor network. Among these, LEACH [4] is one of the most famous clustering protocols. LEACH selects *CH* based on some probability. Due to this, role of *CH* is transferred to other node after every round. The main limitation of LEACH is that low energy sensor node can be selected as *CH*. This causes *CHs* to die quickly. To improve the performance of LEACH, number of clustering protocols has been developed, among these, HEED [5] and PEGASIS [6] are very popular. PEGASIS uses greedy approach to arrange sensor nodes into an ordered list such that each node can communicate with its adjacent nodes in the list. In HEED [5], residual energy of node is the main constrained which is used in the selection of *CH*. This protocol mainly focuses on the energy efficient *CH*s selection and reduction of the communication overhead. This causes to maximize the network lifetime.

In LEACH-C (Centralized LEACH) [7] firstly sink node computes average node energy for each round and nodes having energy higher than average node energy is selected as eligible candidate for becoming *CHs*. Cluster formation is done using simulated annealing-based meta-heuristic algorithm. It performs better than LEACH as it selects *CH* based on energy, thus increases network lifetime. This protocol does not consider, how to balance the

size of cluster which leads to unbalance energy consumption problem. Latiff *et al.* [8] have discussed a PSO-based clustering scheme, called, *PSO-C*. It derives a fitness function which contains two elements such as residual energy, distance between *CH* and *CMs*. It does not consider cluster size which is important factor for reduction of energy consumption. In [9], Kuila *et al*. have proposed a clustering scheme based on differential evolution based optimization algorithm. Lalwani et al. [10] have discussed a BBO-based clustering scheme for selection of *CHs* and cluster formation. The main limitation of the scheme proposed in [9] and [10] are uneven distribution of *CHs* that causes unbalanced energy consumption. In [11], Mann *et al.* have studied artificial bee colony based meta-heuristic algorithm for selection of *CHs* as well as routing of data packet from *CHs* to the *BS*.

In [12], the authors have discussed a PSO-based protocol for Clustered *WSN*, called *E-OEERP*. In *E-OEERP*, PSO based scheme was used for selection of *CHs* and focus on the problem of left-out nodes during clustering process. This scheme is also suffered from unbalanced energy consumption problem. Rao *et al*. [13] have studied a PSO-based cluster head selection method, called *PSO-ECHS*. The main limitation of *PSO-ECHS* is that it does not distribute *CHs* uniformly, thus unbalanced energy consumption was observed during its evaluation. In [14], the authors have studied clustering scheme for heterogeneous *WSN*-based on cuckoo search based meta-heuristic algorithm. This work assumes that 20% nodes have higher energy than the remaining sensor nodes which is an unrealistic assumption. The main drawback of this scheme is that *CHs* are always selected from these higher energy nodes that causes unbalanced energy consumption problem. In order to solve unbalanced energy consumption problem, this paper proposed energy balanced clustering protocol using an improved Cuckoo Search-based meta-heuristic algorithm.

3. Proposed Protocol

In this section, firstly derivation the fitness function for the selection of *CHs* is described and also description of cost function used in the evolution of the cuckoo's nest is discussed. Next, description of the proposed Improved Cuckoo Search-based Clustering Algorithm (*ICSCA*) is presented.

3.1. Derivation of Fitness Function

For selecting eligible *CHs,* fitness function of each node is calculated based on following expression:

$$fitness = a_1 * f_1 + a_2 * f_2 + a_3 * f_3 \tag{1}$$

Where a_1, a_2 and a_3 are constant and its value is between 0 and 1. Value of a_3 equals to (1- a_1-a_2).

$$f_1 = \frac{1}{m} \times \sum_{n=1}^{m} \left[\frac{\sum_{i=1}^{|C_n|} dist(CH_n, CM_i)}{|C_n|} \right] \tag{2}$$

Here, m is number of *CHs,* C_n is number of cluster members in n^{th} cluster and $dist(CH_n, CM_i)$ is distance between n^{th} cluster head and i^{th} cluster member. Second component, f_2, of the fitness function is expressed as follows:

$$f_2 = \frac{E_{residual}}{E_{total}} \tag{3}$$

$E_{residual}$ is the residual energy of node.E_{total} is the total energy of node that are in communication range of particle. Thirst component, f_3, of the fitness function is expressed as follows:

$$f_3 = \frac{C_n}{N} \tag{4}$$

C_n is number of the average cluster member of in a particular cluster. N is total number of nodes. After calculation of fitness function of each node, 20% node is selected that has highest fitness value. From these 20% eligible nodes to be selected as cluster heads, each Cuckoo's nest is populated. This means, these eligible nodes are candidate *CHs, works as Cuckoo's eggs* that is used for the initialization of the hosts' nest. For selection of the best

nest, cost of each host nest is evaluated based on the following cost function:

$$\text{Cost} = \beta * x_1 + (1 + \beta) * x_2 \tag{5}$$

Here sub-components of the cost function (x_1, x_2) is defined as follows:

$$x_1 = \max_{k=1,2,3,\ldots\ldots K} \{\sum d(n_i, CH_{e,k})/|C_{e,k}|\} \tag{6}$$

$$x_2 = \sum_{i=1}^{N} E(n_i) / \sum_{k=1}^{K} E(CH_{e,k}) \tag{7}$$

Here function x_1 denotes maximum average Euclidean distance between nodes (n_i) and their associated *CH*. $C_{e,k}$ denotes number of nodes that are in the communication range of cluster C_k of egg e. Function x_2 denotes ratio of total energy of all nodes to the total energy of all the nodes present in network to total energy of all *CHs,* present in the nest *e*. Value of β is 0.5. Minimum value of Function x_1 and x_2 helps to minimise intra-cluster distance and to select optimum position based *CHs* which reduces energy consumption.

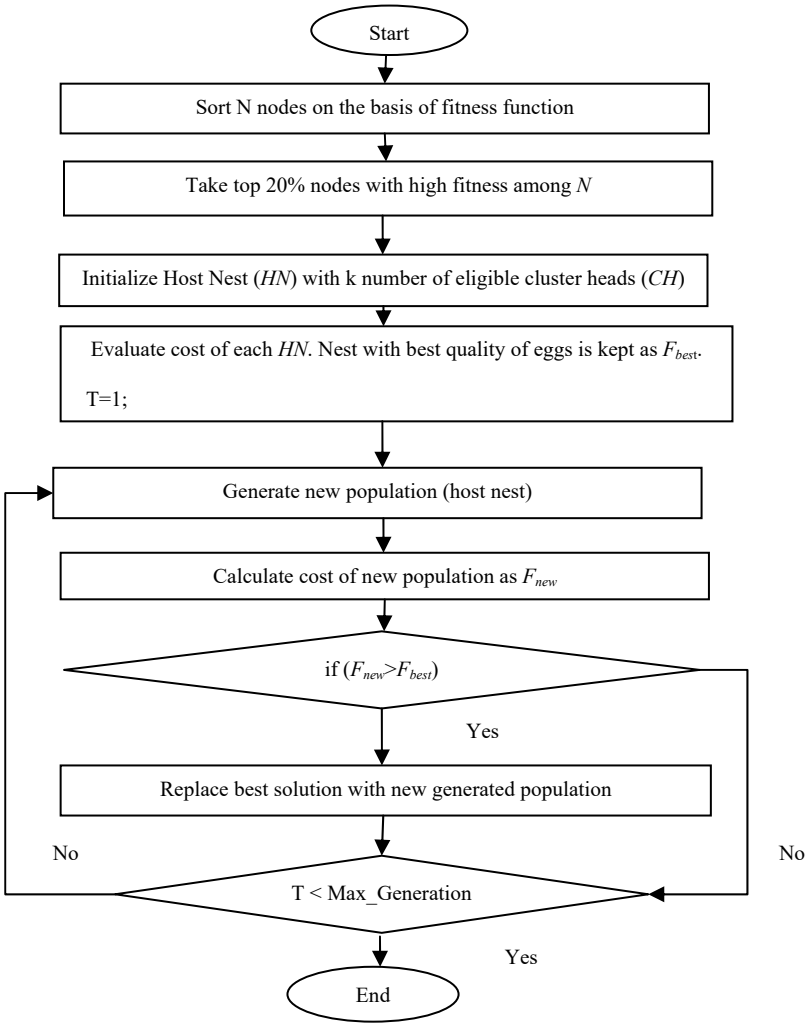


Fig. 1. Flow chart of the working of proposed protocol

3.2. Description of Proposed Clustering Algorithm

This section presents description of the proposed clustering, *ICSCA*, algorithm. The proposed algorithm, *ICSCA* has mainly two components initialization and iterative process of the *ICSCA*. Description of each component is as follows:-

- **Initialization:** In the initialization step, first fitness of each node is evaluated by using Eq. (1) and then sort sensor node list according to its fitness value in descending order. After sorting, top 20% node is selected from sorted list *L*. After this process, initialization of the host's nest with the eggs is done. Here candidate *CHs* presents eggs. These eggs are the sensor nodes randomly selected from sorted list L and each nest populated with *k* number of candidate *CHs*. Here candidate *CH* and sensor nodes are synonymous. We assume *H* is total number of nest and *k* number of eggs. After the population of each nest, cost of each host nest is evaluated using Eq. (5). After this, best nest with highest cost function is selected as a best host nest. This selected best host nest is represented as F_{best}. After this, iterative process of the Cuckoo Search will start for selection of high quality nest which contains best set of the *CHs*.
- **Iterative process of the Cuckoo Search:** This iterative process of the Cuckoo Search algorithm is repeated until stopping criteria is satisfied. In iterative process, first a new Host nest is created with help of *H* host nest which is generated in initialization process. Next, evaluate the cost of the new host nest using Eq. (5), say F_{new}. If value of F_{new} is greater than F_{best}, replace the value of F_{best} by $F_{new.}$ Repeat this process until *Max_Generation* is reached. After the *Max_Generation* iteration, a best nest is obtained which comprises of best set of *CHs*. The flowchart of the proposed improved cuckoo search based clustering algorithm is illustrated in Fig.1.

4. Performance Evaluation

This section presents a performance evaluation of the proposed (*ICSCA*) scheme and its comparison with the three well-known clustering algorithms such as such as *LEACH* [3], *E_OEERP* [12], and *PSO-ECHS* [13]. For the simulation experiments, a custom *WSN* simulator was designed using *MATLAB* where 200 nodes are randomly deployed over a two dimensional area of size 200×200. For energy consumption, we have used same energy model as used in [3]. Simulation parameters are listed in Tabled 1.

Table 1: Parameters list used in Experiments

Parameter	**Value**
Size of Monitoring area	200×200 m^2
Location of the sink	(200-200)
Number of sensor nodes	100-300
Initial Energy of sensor nodes	2J, 200J
% of CHs	5-25
E_{elec}	50 nJ/bit
E_{fs}	10 pJ/bit/m^2
E_{mp}	0.0013 pJ/bit/m^4
d_0	87.00
d_{max}	25m
Packet length	4000bit

Fig.2 (a) illustrates the performance of the *ICSCA* in terms of the total energy consumption with the varying number of rounds. Total energy consumption of the all scheme such as proposed *ICSCA*, LEACH [3], *E_OEERP* [12], and *PSO-ECHS* [13] are estimated with varying number of data collection rounds from 100 to 500. It can be viewed from the Fig.2 (a) that *ICSCA* outperforms the remaining scheme in terms of total energy consumption. This is due to uniform distribution of the *CHs* that improved load of each *CH*.

Fig.2 (b) illustrates the performance of proposed *ICSCA* with existing protocol. In this experiment, area of interest is fixed 200×200. We have varied node density i.e. number of node from 100 to 300. We have calculated total energy consumption for fixed 100 rounds. With increase of node total energy consumption of protocol increases however our proposed protocol performs better than existing protocol.

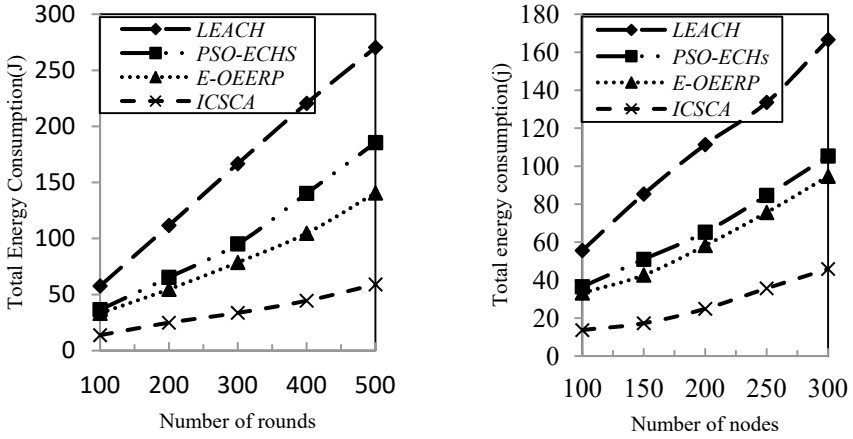


Fig.2. (a) Total Energy consumption vs. Number of rounds (b) Total Energy consumption vs. Number of nodes

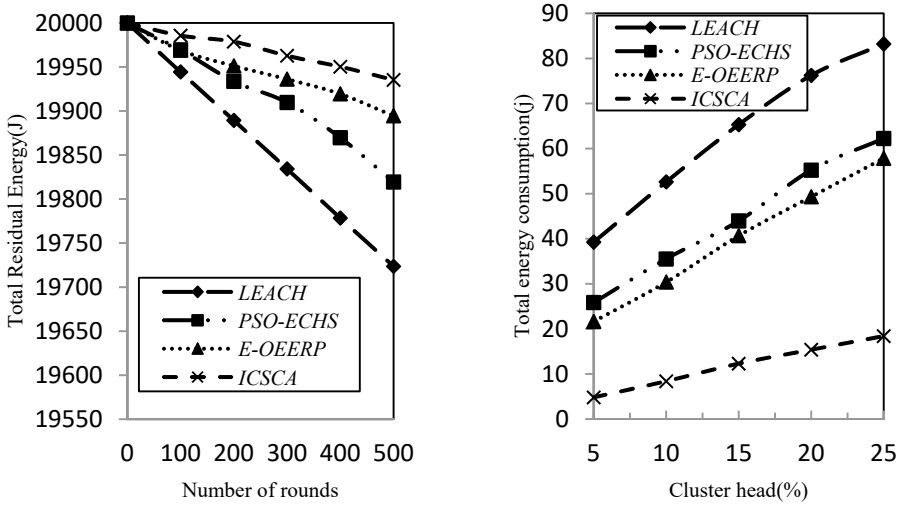


Fig. 3 (a) Total Residual Energy vs. Number of rounds (b) Total Residual Energy vs. % of CHs

Fig. 3(a) illustrates the performance of proposed *ICSCA* scheme and its comparison with *LEACH* [3], *E_OEERP* [12], and *PSO-ECHS* [13] in terms of residual energy. In this experiment, total 200 nodes are deployed in area of interest of dimension 200×200. Nodes are provided with initial energy of 200J. It is observed from Fig. 3(a) that with increase of number of rounds, residual energy of network decreases. However, the proposed protocol performs better than existing protocol in terms of residual energy.

Fig.3 (b) illustrates performance of proposed scheme *ICSCA* and its comparison with *LEACH* [3], *E_OEERP* [12], and *PSO-ECHS* [13] in terms of residual energy by varying the number of *CHs*. In this experiment, area of interest is fixed 200×200. In this experiment, percentage of cluster heads is varied from 5% to 25%. Total number of nodes deployed is 200. With increase of the *CHs*, total energy consumption of the protocol increases. However, proposed

protocol *ICSCA* performs better than existing protocol. This is due to fact that it uses a better fitness function which ensures selected *CHs* are uniformly distributed and their load is balanced. This causes better performance of the proposed scheme.

5. Conclusion

In this work, we have proposed an improved Cuckoo Search based meta-heuristic algorithm for energy balanced clustering. For the energy efficient CHs selection, a novel fitness function is derived. Our proposed clustering algorithm distributes the CHs uniformly so that communication load for them is balanced. In order to transport aggregated data from CHs to the sink, we have used multi-hop routing algorithm for selecting an energy efficient route between CH and BS. Performance comparison of the proposed ICSCA scheme with the other schemes such as LEACH, PSO-ECHS and E-OEERP are discussed. Simulation results confirm that the proposed ICSCA scheme perform well than the other schemes in terms of total energy consumption, residual energy and network life time. This work will be further extended for 3-D wireless sensor networks and its performance may be observed in presence of mobility in the network.

References

[1] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, and E. Cayirci, “Wireless sensor networks: a survey," Computer Networks, vol. 38, pp. 393-422, 2002.
[2] Govind P. Gupta, Manoj Misra, Kumkum Garg, “Energy and trust aware mobile agent migration protocol for data aggregation in wireless sensor networks,” Journal of Network and Computer Applications, Volume 41, May 2014, pp. 300-311.
[3] Gupta, Govind P, “Efficient coverage and connectivity aware data gathering protocol for wireless sensor networks,” Recent Advances in Information Technology (RAIT), 3rd International Conference on. IEEE, 2016.
[4] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, “Energy efficient communication protocol for wireless microsensor networks," in Proc. 33rd Annual Hawaii International Conf. Syst. Sciences, vol. 8, p. 8020, Hawaii, USA, Jan. 2000.
[5] O. Younis, S. Fahmy, “HEED: a hybrid, energy-efficient, distributed clustering approach for ad hoc sensor networks,” IEEE Trans. Mobile Computing. Vol 3(4) , pp. 366–379, 2004.
[6] Adamu Murtala Zungeru, “Classical and swarm intelligence based routing protocols for wireless sensor networks: a survey and comparison,” Journal of Network and Computer Applications, vol. 35, pp. 1508–1536, 2012.
[7] Heinzelman, W. B., Chandrakasan, A. P., & Balakrishnan, H. , “An application specific protocol architecture for wireless microsensor networks,” IEEE Transactions on Wireless Communications, vol. 1(4), pp. 660–670.
[8] Latiff, NM Abdul, Charalampos C. Tsimenidis, and Bayan S. Shari, “Energy-aware clustering for wireless sensor networks using particle swarm optimization.” IEEE 18th International Symposium on Personal, Indoor and Mobile Radio Communications. IEEE, 2007.
[9] Kuila, P., & Jana, P. K., “A novel differential evolution based clustering algorithm for wireless sensor networks,” Applied Soft Computing, 25, 414–425, 2014.
[10] Lalwani, P., Banka, H. & Kumar, C., “BERA: a biogeography-based energy saving routing architecture for wireless sensor networks,” Soft Computing, pp 1–17, 2016.
[11] Mann, P.S. & Singh S., “Artificial bee colony metaheuristic for energy-efficient clustering and routing in wireless sensor networks,” Soft Computing, pp 1–14, 2016.
[12] RejinaParvin, J., and C. Vasanthanayaki, “Particle Swarm Optimization-Based Clustering by Preventing Residual Nodes in Wireless Sensor Networks,” IEEE Sensors Journal, 15.8, pp. 4264-4274, 2015.
[13] Rao, P. C., Jana, P. K., & Banka, H., “A particle swarm optimization based energy efficient cluster head selection algorithm for wireless sensor networks,” Wireless Networks, : 1-16, 2016.
[14] Adnan, Md. Akhtaruzzaman and Razzaque et al., “A Novel Cuckoo Search Based Clustering Algorithm for Wireless Sensor Networks”, Advanced Computer and Communication Engineering Technology: Proceedings of ICOCOE 2015", 2016, Springer International Publishing, 621-634.
[15] Yang, X.-S., & Deb, S., “Multi objective cuckoo search for design optimization,” Computers & Operations Research, 2011.
[16] Valian, E., & Valian, E., “A cuckoo search algorithm by Lévy flights for solving reliability redundancy allocation problems,” Engineering Optimization, 1–14, 2012.
[17] Yang, X.-S., & Deb, S., “Cuckoo search: Recent advances and applications,” Neural Computing and Applications, 1–6, 2013.